Introduction to MS Project

Lab 2 - Programme Monitoring and Control

Table of Contents

1 Introduction 2

1.1 Support Project ..... **Error! Bookmark not defined.**

2 Baseline Programme 3

3 Record Progress 5

4 Earned Value Analysis 8

5 Rescheduling remaining of project 9

6 Conclusion 10

1 Introduction

In **Lab 1**, you have created a complete programme (schedule) for the *planning* of a project. In this lab, you will learn how to use the programme to effectively control the construction project. The lab presents the concept of baseline, and how to use them to compare planned and actual construction performance. The lab will also investigate how to update programmes based on information from the actual construction performance.

To conduct this Lab, please read each main section, looking at the different features in MS Project, but do nothing until you reach the "Apply this" boxed sections. Only do what is suggested in those "Apply this" sections. Do not jump sections or try to do more than what is asked in the given Section; simply follow the document.

2 Baseline Programme

In order to compare a reported progress at any time during delivery against the original plan, we need to set the planned project programme as a *Baseline*. For this, go to the ***Project*** ribbon and click on *Set Baseline...* to open the ***Set Baseline*** window shown in Figure 1. If you haven't set an earlier baseline, set the baseline to *Baseline*; otherwise, you must create a new baseline by select Baseline1 (and so on) in the dropdown menu for *Set baseline*. The baseline is normally set for the entire project, but in some circumstances can also be assigned to keep track of specific tasks since a certain date.

Once you click *OK*, the baseline is saved. After that any change to the programme will be considered by MS Project as newer information that can be compared against the prior one.

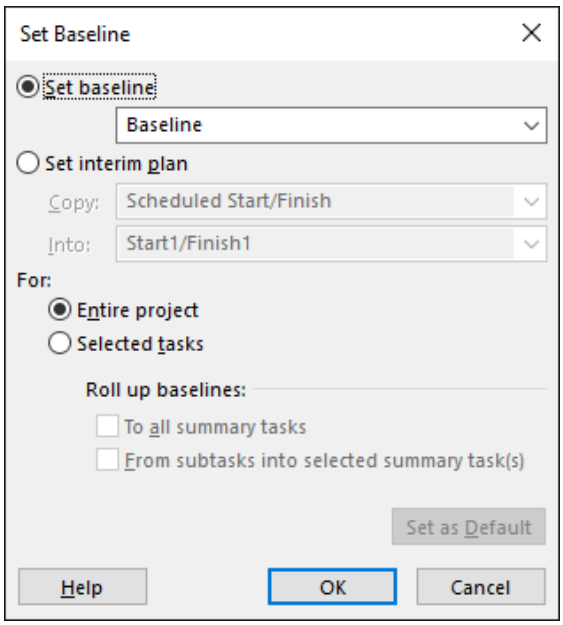


Figure 1: The window for setting a programme as baseline.

3 Status Date

Before entering actual data (e.g. activity progress, actual cost), the date for that actual information must be set. MS Project refers to this date as the *Status Date*. To set it, open the ***Project Information*** window in the ***Project*** ribbon (Figure 2). Note that the status date is set to NA by default. should probably be the same as the current date unless it is felt otherwise. Note that the status date can also be changed directly in the *Status* section of the ***Project*** ribbon by clicking on the little calendar icon next to *Status Date*.

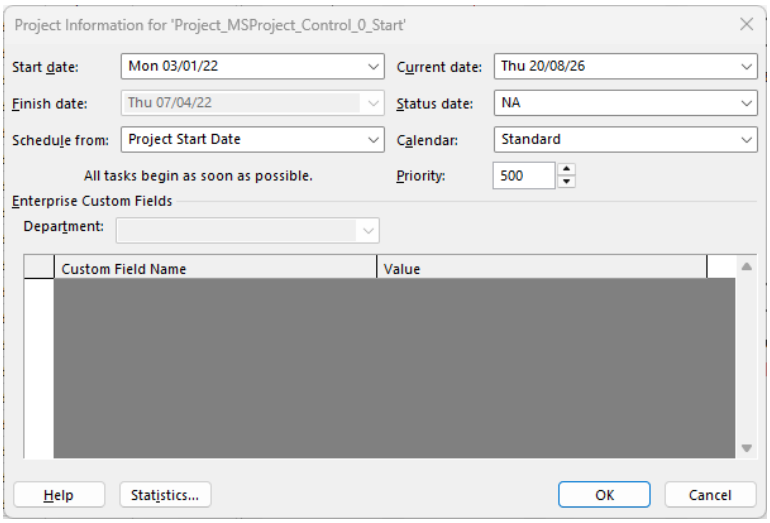


Figure 2: Project Information window

Apply this:

1. Define your existing programme as baseline.

Using the project information in the Support Project sheet and the progress information in Section **Error! Reference source not found.**:

2. Set the Status Date.

4 Record Progress

After setting a baseline and a status date we can enter actual progress information up to that status date. To do this, go to the ***Gantt Chart*** view and change the table format to *Tracking*. This table provides the means to record:

- *% Complete*:

$$\% \text{ Complete } = \frac{\text{Actual Duration}}{\text{Duration}} \times 100$$

  If the *% Complete* field is set to a value greater than zero, the *Actual Start* field is set to the scheduled start date, unless you have manually entered a different actual start date. If the *% Complete* field is set to 100, the *Actual Finish* field is set to the scheduled finish date. If you type a value in the *% Complete* field, MS Project automatically calculates actual duration and remaining duration. Likewise, entering a value in the *Actual Duration* or *Remaining Duration* field automatically recalculates the other fields.

- *Physical % Complete:* This field can be used when the calculated percent complete would not be an accurate measure of real work performed or measured. Unlike the *% Complete* field, the *Physical % Complete* field is independent of the total duration or actual duration values used by the *% Complete* field to calculate BCWP. So, do not use it to calculate EVA performance metrics.

- *Actual Start/Finish*: These are automatically calculated, unless they are different from the baseline values. If the start of an activity is delayed that you need to enter a specific *Actual Start* value manually. The same applies to *Actual Finish*.

- *Actual Duration*: This has the *Baseline* value by default, but it can be altered manually.

- *Remaining Duration*: This is calculated automatically based on the *Actual Duration* and the reported progress.

- *Actual Cost*: This is calculated automatically based on the *Actual Work* (see below) and the actual Costs (that can be changed on the ***Resource Sheet*** view). But, you can choose to turn off automatic calculation of costs though, so that you can enter exact actual values (which is needed to calculate CPI or SPI, for example). To do this, go to ***File***→*Options*→*Schedule* and go to the section *Calculation options for this project*. There, clear the checkbox *Actual costs are always calculated by Project*. (do not tick the enabled sub-tickbox *Edits to total actual cost will be spread to the status date*).

- *Actual Work*: This is linked to *Actual Duration*.

IMPORTANT NOTE: When you enter an *Actual Start* value manually, then MS Project by default updates the Gantt chart automatically. On the other hand, when you enter a progress value that leads to an activity being behind schedule, the schedule does not update itself automatically. Indeed, there are many options to deal with such situation, and so MS Project does not make a default decision for you. If you wish to maintain the rate of work and thus extend the activity duration in accordance to the delay, you can use the *Reschedule uncompleted work* function mentioned above and detailed in **Section 6**.

Another useful tool is the set of *Tracking* options in the *Schedule* section of the ***Task*** ribbon. Shown in Figure 3 the section includes five convenient buttons to set progress to 0%, 25%, 50%, 75% and 100%. The button *Mark on Track* enables to set the activity exactly on track, i.e. for which the progress up to the status date is as planned in the baseline programme.

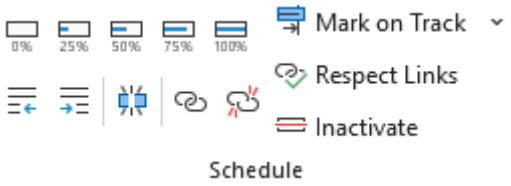


Figure 3: The *Schedule* section of the *Task* ribbon.

Alternatively, there is *Update Project* button in the ***Project*** ribbon (Figure 4) that opens the ***Update Project*** window shown in Figure 5. This window enables the user to:

1) automatically *Update work as complete* (on track) based on the status date. This is equivalent to applying the *Mark on Track* setting above to all tasks, which can speed up the process of entering progress information; and subsequently
2) *Reschedule uncompleted work* (see **Section 6**).

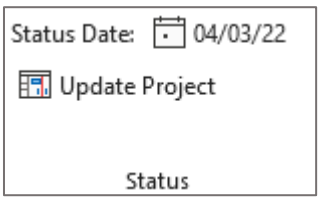


Figure 4: Status section of the Project ribbon.

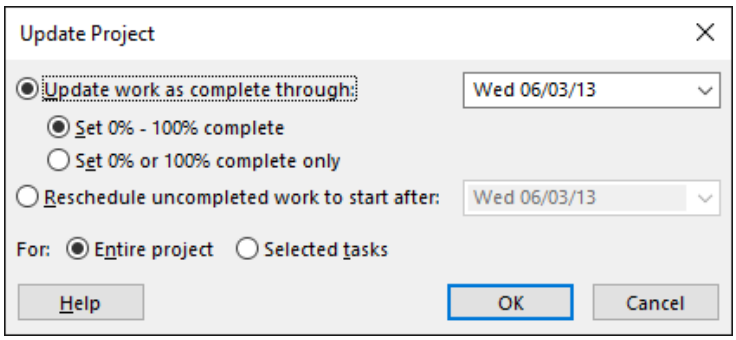


Figure 5: The *Update Project* window.

Progress is displayed graphically in the ***Tracking Gantt*** view (additionally, it is good to have the table format set to *Tracking*). You can also show a *Progress Line* in the Gantt chart view by right-clicking on the Gantt chart area and select *Progress Lines*. This opens the ***Progress Lines*** window in Figure 6 where you simply need to display the *Current progress line At project status date*.

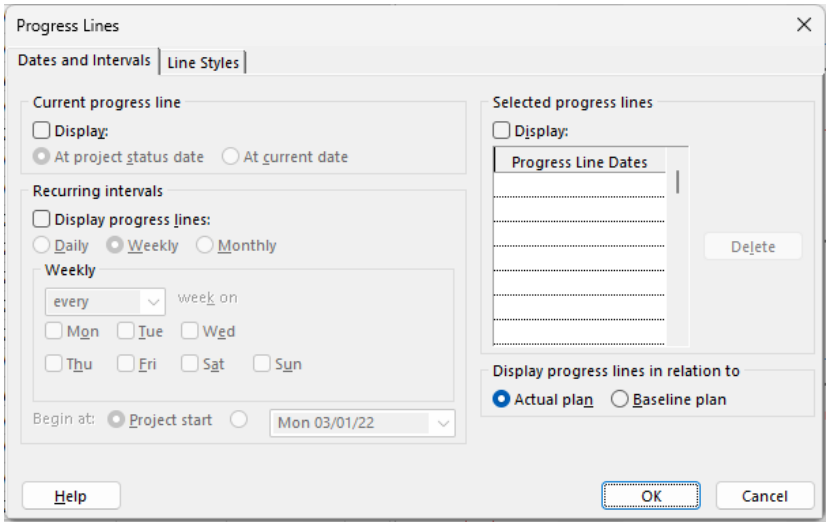


Figure 6: *Progress Lines* window.

IMPORTANT NOTE: It is important to understand that the resource levelling previously done is discarded by MS Project when updating a schedule with actual information. Therefore, you may need to re-level your schedule after entering the actual progress information.

Apply this:

Using the Project Monitoring Stage information in the House Project document:

1. Enter progress information for all activities.
2. Open the *Tracking Gantt* view and show the *Progress Line* at the project status date.

5 Earned Value Analysis

After entering the actual progress and actual costs of completed work, we can compare them against the baseline (i.e. scheduled, or planned) one, to infer project performance in terms of time and cost through the Earned Value indicators.

To do this, change to the table format to *Earned Value*. Note that this is not one of the standard table formats. Therefore you need to select it from the complete list of tables available under *More Tables...* . The Earned Value table is as expected and can be used to compare the earned value with the planned value. The table contains the following columns

- *Planned Value - PV (BCWS):* Budgeted (planned) cost for the work originally scheduled (planned) up to the status date.
- *Earned Value - EV (BCWP):* Budgeted (planned) cost for the work actually performed (progress).
- *Actual Cost (ACWP):* Actual cost for the work actually performed (progress).
- *Schedule variance (SV)*: the difference between the current progress and the scheduled progress of a task, in terms of cost (the formula SV = Planned Value – Earned Value).
- *Cost variance (CV)*: the difference between a task's estimated cost and its actual cost (the formula CV = Earned Value – Actual Cost).
- *Estimate at completion (EAC)*: is the estimated total cost of a task or project, based on performance as of the status date.
- *Budget at completion (BAC)*: shows an estimate of the total project cost.
- *Variance at completion (VAC)*: shows the difference between the budget at completion (BAC) and the estimate at completion (EAC).

One can also add the *CPI* and *SPI* columns to the table. They are already calculated by MS Project but are not automatically displayed in this table. To show them, right click anywhere on the table header row and select *Insert Column*.

While the above table gives information in a WBS-aggregation style, one may also be interested in visualising the BCWP (Earned Value), BCWS (Planned Value) and ACWP (Actual Value) over time. This can be done through the *Earned Value* report available among the *Costs* reports in the ***Report*** ribbon. A similar *Visual Report* is also available.

Apply this:

Produce an *Earned Value* report for your project. In the report, change the *Indices over Time* diagram to instead show the CV and SV per WBS task at level 2. Which task(s) is(are) experiencing difficulties?

6 Rescheduling remaining of project

The Earned Value Analysis (EVA) uses only information about the planned and actual programme up to the status date. The work conducted above thus does not address the impact of changes to date (status date) onto the remaining of the project.

Rescheduling the remaining of the project can be done manually by increasing the duration of activities or moving them. Note, though, that MS Project includes some functionality to do this automatically. For this, go to the ***Project*** ribbon and open again the *Update Project* window (see Figure 4 and Figure 5). This time, select the second option bullet point *Reschedule Uncompleted Tasks to Start after*. The date should be set to the Status Date. After clicking *OK*, MS Project will reschedule the activities to address a delay or take advantage of any work ahead of schedule.

Apply this:

1. Reschedule your project to take into account the latest progress information. Has the completion date changed?

7 Conclusion

This is the end of our labs on Programming and Monitoring with MS Project. There are many other software packages available for (construction) project programming and monitoring, such as Primavera P3. One cannot presume of the software that the company they will work for will use, but all of them have the core functionalities presented in these labs.

Additionally, MS Project has many more functionalities that have not been investigated here, in particular some for collaborative working. The Internet is full of useful tutorials and discussion forums about MS Project (starting with the official Microsoft Project website) through which you can discover all those functionalities.